

NIHAL

Senior Solution Architect

PROFESSIONAL SUMMARY

Senior Solution Architect with 8+ years of professional software engineering and enterprise architecture experience, including over 5 years specializing in the Insurance, Membership, and Payment domains. Expert in designing end-to-end enterprise solutions, microservices architectures, API integrations, cloud platforms (GCP and Azure), and modern DevOps pipelines. Strong hands-on technical proficiency with Python development, RESTful APIs, event-driven architectures, Docker, Kubernetes, and secure enterprise integrations (OAuth 2.0, JWT, SAML). Proven track record of leading 9 comprehensive enterprise projects, ensuring strict adherence to enterprise security policies, scalability, and disaster recovery standards.

CORE SKILLS

Category	Skills & Technologies
Solution Architecture	Enterprise Solution Design, Application & Integration Architecture, Microservices, Event-Driven Architecture, API Strategies, Disaster Recovery Planning
Core Programming & APIs	Python (8+ years), RESTful APIs, API Gateways, OAuth 2.0, JWT, SAML, Enterprise Integration Patterns
Cloud Platforms & DevOps	Google Cloud Platform (GCP), Azure, Docker, Kubernetes, CI/CD Pipelines, Infrastructure as Code (IaC), Git, Terraform
Databases & Messaging	Relational & NoSQL Databases, PostgreSQL, MongoDB, Apache Kafka, RabbitMQ, Azure Service Bus

Enterprise Domains & Governance	Insurance, Membership, and Payment Domains, Agile/Scrum Methodologies, Stakeholder Management, Architecture Governance
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PROFESSIONAL EXPERIENCE

Project 1: Enterprise Insurance & Membership Policy Engine

- **Role:** Lead Solution Architect
 - **Duration:** Aug 2025 – Present
 - **Tech Stack:** Python, Microservices, GCP, Kubernetes, Docker, Apache Kafka, PostgreSQL, OAuth 2.0, REST APIs
 - **Project Description:**
 - Designed end-to-end solution architecture for high-scale insurance policy lifecycle and membership management platforms aligned with business and technical requirements.
 - Defined application, integration, data, and security architectures utilizing microservices and containerized GCP environments.
 - Led architecture discussions and provided technical direction for real-time event streaming workflows powered by Apache Kafka and Python microservices.
 - Ensured solutions adhered strictly to enterprise security policies, data protection guidelines, and OAuth 2.0/JWT authentication standards.
 - Drove performance optimization, high availability, and disaster recovery planning for mission-critical core insurance transaction systems.
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Project 2: Secure Payment Gateway & Transaction Processing Microservices

- **Role:** Senior Solution & Integration Architect
- **Duration:** Jun 2024 – Jul 2025
- **Tech Stack:** Python, FastAPI, RabbitMQ, GCP, Docker, Kubernetes, Redis, MongoDB, CI/CD
- **Project Description:**
 - Spearheaded the architecture and development of scalable payment gateway microservices handling multi-channel merchant transactions securely.
 - Defined integration patterns, API strategies, and secure event-driven messaging layers using RabbitMQ and Python.
 - Reviewed and approved solution designs, technical specifications, and implementation approaches across distributed engineering squads.
 - Provided technical leadership to development teams during implementation, deployment, and cloud-native container orchestration via Kubernetes.
 - Partnered with security stakeholders to identify technical risks, dependencies, and robust mitigation strategies for financial data pipelines.

Project 3: Automated Membership Billing & Subscription Hub

- **Role:** Senior Enterprise Architect
- **Duration:** Apr 2023 – May 2024
- **Tech Stack:** Python, Django, PostgreSQL, Azure Service Bus, Docker, Kubernetes, Jenkins, Git
- **Project Description:**
 - Directed the architectural modernization of membership subscription billing engines supporting millions of active global users.
 - Defined data governance, metadata management, and secure API integration standards across hybrid cloud infrastructure.
 - Configured comprehensive integration patterns using Azure Service Bus and asynchronous Python worker queues.
 - Established automated testing frameworks and continuous integration pipelines (CI/CD) to streamline deployment velocity.
 - Led technical architecture reviews and cross-functional workshops with product owners and business stakeholders.

Project 4: Cloud-Native Insurance Claims Analytics & Decision Support

- **Role:** Senior Cloud Solution Architect
- **Duration:** Mar 2022 – Mar 2023
- **Tech Stack:** Python, GCP, BigQuery, REST APIs, Docker, Terraform, Kubernetes
- **Project Description:**
 - Designed curated data products and semantic layers on Google Cloud Platform to empower business users with automated claims processing intelligence.
 - Leveraged GCP cloud infrastructure to build scalable Python-based backend decision-support microservices.
 - Ensured application compliance with financial security policies and coordinated vulnerability remediation tracking prior to production deployment.
 - Maintained test documentation, architecture blueprints, and API gateway configurations to satisfy industry compliance standards.
 - Supported DevOps, cloud migration, and infrastructure-as-code (IaC) automation initiatives using Terraform.

Project 5: Real-Time Payment Fraud Detection & Ingestion Engine

- **Role:** Senior Data & Integration Architect
- **Duration:** Apr 2021 – Feb 2022
- **Tech Stack:** Python, Apache Kafka, PostgreSQL, Redis, Docker, Kubernetes, Prometheus, Grafana
- **Project Description:**
 - Designed resilient streaming ingestion frameworks to process massive high-frequency financial transaction feeds in real-time.

- Implemented automated validation scripts and fraud scoring logic using Python microservices and distributed caching layers.
- Optimized container resource allocation, auto-scaling rules, and cluster compute policies to minimize latency and infrastructure overhead.
- Authored detailed architecture diagrams and technical documentation for internal development teams and security audits.
- Maintained version control for infrastructure-as-code and service definitions using Git and automated deployment workflows.

Project 6: Containerized Membership Portal & API Gateway Platform

- **Role:** Senior Platform Architecture Lead
- **Duration:** Jun 2020 – Mar 2021
- **Tech Stack:** Python, Flask, Docker, Kubernetes, Kong API Gateway, PostgreSQL, RabbitMQ
- **Project Description:**
 - Built and optimized containerized microservices interacting with Kong API Gateways and centralized relational databases.
 - Validated complex data transformations and RESTful API data flows between core enterprise systems and third-party vendor applications.
 - Configured performance monitoring metrics and health checks using modern observability tools during high-concurrency stress testing phases.
 - Resolved critical architectural defects and coordinated emergency patch deployments during high-priority production support cycles.
 - Standardized data ingestion workflows to support parallel processing streams across distributed engineering teams.

Project 7: Legacy Insurance System Migration & API Harmonization

- **Role:** Solution Architecture & Migration Specialist
- **Duration:** Aug 2019 – May 2020
- **Tech Stack:** Python, SQL, Oracle, REST APIs, Git, Linux, Docker
- **Project Description:**
 - Performed systematic analysis and harmonization of legacy insurance databases and monolithic backend systems.
 - Wrote automated validation scripts in Python and SQL to verify data consistency across legacy schemas and modern microservices.
 - Tracked deployment dependencies and assisted in verifying environment readiness from staging to production cloud zones.
 - Administered bug tracking repositories and organized historical architecture documentation libraries.
 - Maintained technical log documentation and platform status registers for executive project governance audits.

Project 8: Procurement & Supply Chain Financial Ledger Architecture

- **Role:** Systems & Integration Architect
 - **Duration:** Dec 2018 – Jul 2019
 - **Tech Stack:** Python, SQL Server, Shell Scripting, Linux, REST APIs, Git
 - **Project Description:**
 - Assisted in analyzing legacy financial ledgers, stored procedures, and ERP integration scripts for future cloud migration readiness.
 - Supported data profiling tasks, error log collection, and routine API specification documentation protocols.
 - Implemented small-scale Python automation scripts for recurring staging environment health reporting and data reconciliation.
 - Participated in initial database diagnostic monitoring and query optimization exercises under senior guidance.
 - Documented operational configuration steps and support ticket resolutions for internal reference.
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Project 9: Enterprise Data Onboarding & Initial Utility Validation

- **Role:** Junior Software & Architecture Support Engineer
 - **Duration:** May 2018 – Nov 2018
 - **Tech Stack:** Python, Linux, SQL, Basic Scripting, Excel
 - **Project Description:**
 - Assisted senior solution architects with baseline data profiling, database ingestion verification, and metric validation tasks.
 - Maintained local configuration logs and basic intake validation check scripts for structured data files.
 - Supported team documentation readiness and asset register maintenance.
 - Participated in early-stage architectural tool evaluations and operational review meetings.
 - Organized baseline technical reference guides and environment setup logs.
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EDUCATION

Diploma in Computer Science Engineering – Govt. Kalaniketan Polytechnic College, Jabalpur
Passing Year: 2018